

Lightweight Skin Lesion Segmentation with Depth- wise Asymmetric Bottleneck and Marginal Dice Loss

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<https://github.com/anhlethicham-source/LSNet-DAB>

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Abstract

Accurate segmentation of skin lesions from dermoscopic images is a clinically consequential task, yet existing lightweight models either rely on specialised GPU kernels incompatible with standard clinical hardware or fail to address the dual-scale challenge intrinsic to dermoscopic imagery - where diagnostically critical boundary detail co-exists with large, morphologically irregular lesion extents. This paper presents DaLS Net (Depth-wise Asymmetric Large-Small Network), a lightweight hybrid CNN-Transformer framework that addresses both constraints simultaneously. The principal contribution is the Depth-wise Asymmetric (Da) block, a hardware-agnostic convolutional mixer that factorises multi-scale context aggregation into two parallel asymmetric depthwise branches: a Large-kernel path (DaL, $K_L = 7$) that encodes coarse structural context via non-dilated and dilated $7 \times 1/1 \times 7$ sub-paths, and a Small-kernel path (DaS, $K_S = 3$) that preserves fine-grained boundary detail through three complementary $3 \times 1/1 \times 3$ dilated sub-paths. A per-stage dilation schedule - fixed $d = 2$ at high-resolution stages and cycling $d \in \{4, 4, 8, 8, 16, 16\}$ at the deepest convolutional stage - progressively expands effective receptive field coverage without increasing parameters. Global self-attention is confined to the stride-64 bottleneck stage via the MSA block (Wang et al., 2025), where quadratic complexity is tractable and lesion-level relational reasoning is most consequential. The encoder is coupled with a Semantic FPN neck and a lightweight additive FPNHead decoder. Training employs Marginal Dice Loss (Nguyen et al., 2025), which assigns spatially weighted supervision to three disjoint pixel zones - boundary margin ($w = 6$), lesion interior ($w = 3$), and background ($w = 1$) - jointly resolving class imbalance and boundary ambiguity. Evaluated on the ISIC 2018 benchmark, DaLS Net achieves a Dice Similarity Coefficient of 90.04%, Precision of 90.90%, and F1-score of 90.04%, competitive with Transformer- and state-space-based segmenters while requiring no specialised GPU kernel support. These results demonstrate that principled lightweight design, combined with boundary-aware supervision, offers a viable path toward accurate and deployable medical image segmentation.

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1 Introduction

Skin and subcutaneous diseases (SSDs), particularly malignant melanoma, impose a severe global health burden, with incidence rising by over 35% since 1990 (Li *et al.*, 2025). Because melanoma survival rates plummet from 99.5% for localised cases to 31.9% upon distant metastasis (Young *et al.*, 2023), early detection is critical. However, specialist shortages frequently hinder timely diagnosis (Johnson *et al.*, 2025). Deep learning offers a scalable solution; notably, Convolutional Neural Networks (CNNs) have achieved diagnostic specificities (82.5%) that surpass those of level-I dermatologists (Haenssle *et al.*, 2018).

Within the diagnostic pipeline, pixel-level segmentation is fundamentally more complex than simple image-level classification. By accurately delineating anatomically meaningful boundaries, precise segmentation is foundational for clinical workflows, enabling reliable size measurement, longitudinal progression tracking, and targeted biopsy guidance.

Skin lesion segmentation in dermoscopic imagery presents unique technical challenges compared to natural image processing. Key difficulties include ill-defined boundaries with low peripheral contrast, extreme morphological scale variability requiring multi-dimensional spatial reasoning, and confounding artifacts (e.g., hair occlusion, specular reflections) that obscure local textures. Additionally, severe foreground-background class imbalance and inter-patient skin tone variations frequently bias predictions toward the background.

To address these challenges, Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) serve as the dominant foundational architectures. While CNN-based models, such as U-Net, excel at local feature extraction, their restricted receptive fields inherently limit the modeling of long-range dependencies. Conversely, ViTs effectively capture global context but demand prohibitive computational resources, impeding their real-world clinical deployment. Consequently, there is an urgent need for lightweight models capable of balancing segmentation accuracy with computational efficiency.

Recently, lightweight deep learning models have been introduced to address the trade-off between segmentation accuracy and computational efficiency, with LSNet (Wang *et al.*, 2025) being a representative example. However, the characteristic convolution module of LSNet (LSConv) still presents certain limitations: it relies on spatial kernels that require specialized hardware support and has not been fully optimized for extracting fine-grained details in blurred boundary regions of medical images. In addition, training with

conventional loss functions such as Binary Cross-Entropy is often dominated by the excessively large background region, causing the network to overlook prediction errors around lesion boundaries.

To comprehensively address the aforementioned limitations, this study proposes an improved medical image segmentation framework named DaLS Net. The core contributions of this report are reflected in two main aspects:

- We propose **DaLS Net**, a novel lightweight medical image segmentation framework combining a Depth-wise Asymmetric Bottleneck backbone with a Semantic FPN neck and a lightweight Head-Decode .
- We introduce the **Depth-wise Asymmetric Bottleneck Large-Small (DaLS) block**, a hardware-agnostic convolutional mixer that factorises multi-scale context aggregation into parallel large-kernel (DaL) and small-kernel (DaS) asymmetric depthwise branches, each augmented with dilated sub-paths, enabling wide receptive field coverage without specialised GPU kernel support.
- We integrate **Marginal Dice Loss** (Nguyen et al., 2025) into the DaLS Net training pipeline, combining spatially-weighted boundary supervision with a global Dice term to jointly address class imbalance and boundary ambiguity - a combination not previously explored with this class of lightweight convolutional backbone.
- Extensive experiments on the ISIC 2018 benchmark, including ablation studies comparing the Da block against alternative mixer designs, demonstrate that DaLS Net achieves competitive accuracy relative to state-of-the-art methods at significantly lower computational cost.

2 Literature Review

2.1 Deep Learning Architectures for Skin Lesion Segmentation

U-Net (Ronneberger et al., 2015) established the encoder–decoder paradigm that dominates medical image segmentation, achieving precise boundary delineation through skip connections that recover spatial detail lost during downsampling. U-Net++ (Zhou et al., 2018) and Attention U-Net (Oktay et al., 2018) progressively refined boundary localisation through dense nested pathways and soft attention gates, respectively. Despite their effectiveness, these CNN-based architectures rely on fixed-size convolutional kernels, imposing a hard ceiling on receptive field size that constrains modelling of large,

irregular lesions.

Vision Transformers (ViTs; Dosovitskiy et al., 2021) address this limitation by modelling global context through self-attention over patch sequences. Trans-UNet (Chen et al., 2021) demonstrated their utility in medical segmentation by combining CNN feature extraction with Transformer-based global reasoning; however, the quadratic self-attention complexity renders these models prohibitively expensive at dermoscopic resolutions and poorly suited to resource-constrained clinical environments. Recent state-space models - U-Mamba (Ma et al., 2024) and VM-UNetV2 (Zhang et al., 2024) - offer linear-complexity long-range modelling, but process 2D feature maps as flattened 1D sequences, inherently disrupting the spatial continuity that precise boundary delineation requires. These limitations collectively motivate a purely convolutional, hardware-agnostic architecture that achieves wide receptive field coverage without the overhead of attention mechanisms or specialised GPU kernels.

2.2 Lightweight Convolutional Designs and Asymmetric Convolutions

The efficiency frontier in medical image segmentation has been progressively advanced by depthwise separable convolutions (Howard et al., 2017), which decouple spatial filtering from channel mixing to reduce per-layer FLOPs from $O(K^2C^2)$ to $O(K^2C + C^2)$. U-Lite (Dinh et al., 2023) further exploited this principle through Axial Depthwise Convolution - combining $1 \times n$ and $n \times 1$ depthwise kernels additively with a residual connection - achieving under 0.9M parameters while attaining 90.39% DSC on ISIC 2018. This demonstrates that large receptive fields can be approximated through sequential 1D operations, a key insight for resource-constrained deployment. DABNet (Li et al., 2019) formalised the asymmetric factorisation principle: decomposing a $K \times K$ depthwise convolution into sequential $K \times 1$ and $1 \times K$ kernels reduces per-pixel cost from $O(K^2)$ to $O(K)$, and pairing this with a dilated branch at rate d expands the effective receptive field to $(6d+1) \times (6d+1)$ at zero additional parameter cost. Its cycling dilation schedule — applying progressively increasing rates across successive blocks — empirically outperforms both constant-rate and coprime alternatives for multi-scale feature extraction. However, DABNet’s $3 \times 1 / 1 \times 3$ base kernels cover only a 3×3 effective receptive field in the non-dilated path, insufficient to span the wide spatial extents of large dermoscopic lesions that routinely exceed 100×100 pixels.

DaLS Net addresses this gap by widening the asymmetric kernels to $7 \times 1 / 1 \times 7$, recovering a 7×7 base receptive field in the non-dilated path. The Da block further fuses a Dual-Asymmetric Large-kernel (DaL) branch and a Dual-Asymmetric Small-kernel (DaS) branch in parallel: the DaL branch captures broad contextual structure across large lesion extents, while the DaS branch preserves the fine-grained local detail - such as the narrow rim of boundary pixels - that wider kernels tend to smooth over. This parallel dual-branch design is the principal architectural contribution of DaLS Net and is described in full in Section 3.3.

2.3 Feature Pyramid Networks and Boundary-Aware Loss Functions

Feature Pyramid Networks (Kirillov et al., 2019) aggregate multi-scale backbone features via a top-down lateral pathway, ensuring small lesions benefit from high-resolution feature maps while large lesions access semantically rich deep representations simultaneously. DaLS Net adopts the Semantic FPN paradigm without modification, as the four-level feature hierarchy of the DaLS encoder provides a natural fit for FPN-based aggregation at low parameter overhead.

Standard Dice loss handles class imbalance effectively but assigns uniform weight to all lesion pixels, providing no incentive to prioritise the boundary zone where clinical segmentation errors are most consequential. Tran et al. (2024) introduced Marginal Weight Loss, deriving per-pixel weights for three disjoint spatial zones - background, lesion interior, and boundary margin - with the highest penalty assigned to the margin zone. (Nguyen et al., 2025) extended this into Marginal Dice Loss (MDL) by integrating a global Dice term, jointly resolving class imbalance and boundary ambiguity. MDL has been validated exclusively on heavyweight Mamba- and Transformer-based backbones; its complementarity with a lightweight convolutional backbone of the kind proposed here has not been previously explored.

3 Methodology

3.1 Architecture Overview

We propose **DaLS** (Depth-wise Asymmetric Large-Small), a lightweight hybrid CNN-Transformer network for binary skin lesion segmentation on the ISIC 2018 dataset. The overall pipeline follows an encoder-neck-decoder design, as illustrated in Figure 1. Given a dermoscopic input $X \in \mathbb{R}^{3 \times H \times W}$, the DaLS encoder extracts hierarchical features at four spatial scales through

four successive stages; a Semantic FPN neck fuses these multi-scale representations via a top-down pathway; and an FPNHead decoder produces the final single-channel logit map, converted to a binary mask by sigmoid thresholding at 0.50.

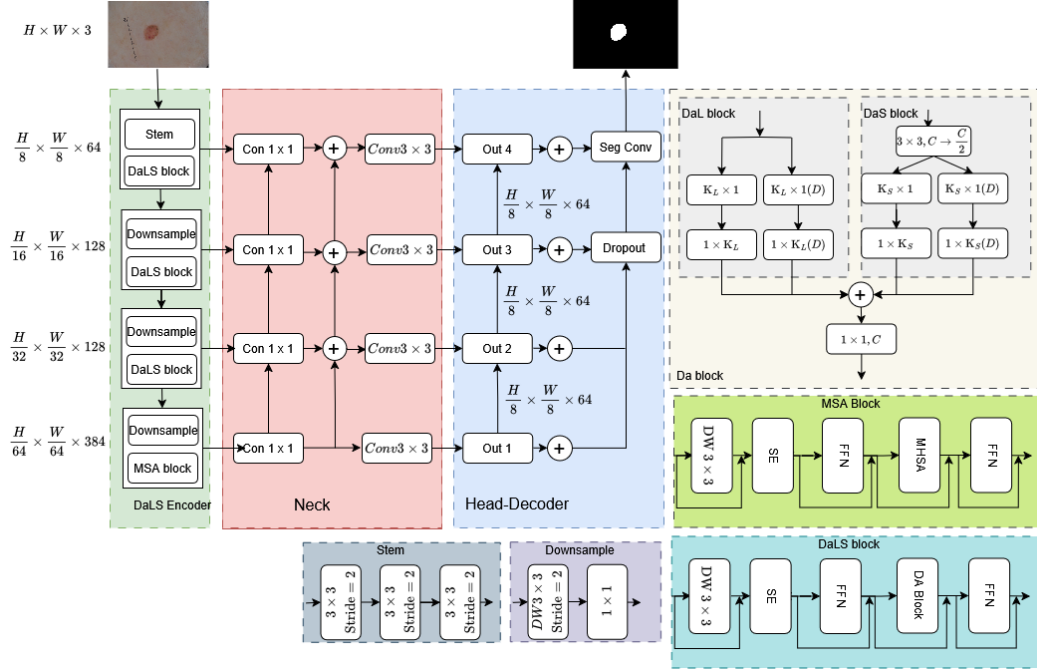


Figure 1: Overall architecture of the proposed DaLS Net segmentation framework.

The design philosophy of DaLS addresses three limitations common to existing lightweight segmentation backbones. First, symmetric encoder-decoder architectures (Ronneberger et al., 2015; Zhou et al., 2018) incur substantial parameter overhead through transposed convolution stacks and dense skip connections, which is inconsistent with efficient inference; DaLS instead employs a single-pass Semantic FPN that recovers spatial detail without a symmetric decoding branch. Second, standard convolutions and self-attention each mix perception and aggregation at a single fixed scale, constraining either receptive field coverage or computational efficiency; DaLS decouples these through the proposed DA block, which handles multi-scale context at low cost. Third, global attention is tractable only at low spatial resolution; DaLS confines it to the deepest stage where both the sequence

length and the semantic abstraction make it most effective.

3.2 DaLS Encoder

The encoder consists of a stem followed by four stages operating at spatial strides $\{8, 16, 32, 64\}$ with channel widths $\{64, 128, 128, 384\}$. The stem reduces the input to $\frac{H}{8} \times \frac{W}{8}$ via three successive 3×3 convolutions with strides $\{2, 2, 2\}$, providing overlapping patch embedding that preserves more spatial detail than non-overlapping tokenisation (Xiao et al., 2021). Inter-stage downsampling uses a strided depthwise 3×3 convolution followed by a pointwise 1×1 projection, decoupling spatial reduction from channel mixing and avoiding the checkerboard artefacts associated with strided pointwise operations. Stages 0-2 consist of stacked **DaLS blocks**. Each DaLS block applies the following sequential computation:

$$u_1 = \text{DW}_{3 \times 3}(x) + x, \quad u_2 = \text{SE}(u_1), \quad (1)$$

$$u_3 = \text{FFN}(u_2) + u_2, \quad u_4 = \text{Da}(u_3) + u_3, \quad (2)$$

$$y = \text{FFN}(u_4) + u_4 \quad (3)$$

where $\text{DW}_{3 \times 3}$ is a depthwise 3×3 convolution for fine-grained local feature extraction, SE is Squeeze-and-Excitation channel recalibration (Hu et al., 2018) that adaptively reweights channels based on global average-pooled statistics, and the two FFN modules are two-layer pointwise expand-contract MLPs with GELU activation and an expansion ratio of 4. Da denotes the Depth-wise Asymmetric block detailed in Section 3.2.1. Independent residual connections wrap each sub-module, preserving gradient flow across the full block depth and enabling the early layers to learn local texture representations before multi-scale context is injected by the Da block. Stage 3 replaces the DaLS block with the MSA block described in Section 3.2.2.

3.2.1 Depth-wise Asymmetric Large-Small (DaLS) Block

The DaLS block is the principal architectural contribution of this work, designed to aggregate multi-scale spatial context in a hardware-agnostic, parameter-efficient manner suited to the heterogeneous morphology of dermoscopic lesions. Dermoscopic images present a dual-scale challenge: lesions span large, irregular extents requiring wide contextual coverage, while diagnostically relevant features such as irregular pigment networks and atypical vascular patterns are concentrated at fine spatial scales near the lesion boundary. A single-scale mixer cannot address both simultaneously without

prohibitive parameter cost. **Overall DaLS block structure.** The DaLS block follows the macro-design of the LS block (Wang et al., 2025), arranging four sequential components around a long residual connection:

$$\text{DaLS}(x) = x + \text{FFN}(\text{Da}(\text{FFN}(\text{SE}(\text{DW}_{3 \times 3}(x))))), \quad (4)$$

where $\text{DW}_{3 \times 3}$ is a depthwise 3×3 convolution that injects local structural priors, SE is a Squeeze-and-Excitation channel-attention layer (Hu et al., 2018) for inter-channel recalibration, the two FFN modules perform point-wise channel mixing before and after the spatial mixer, and Da is the Depth-wise Asymmetric Block that constitutes the core spatial mixing operation. This arrangement mirrors the LS block template while replacing LS convolution with the domain-specific Da, preserving the proven macro-structure without reintroducing the dynamic-weight overhead of the original small-kernel aggregation step. GroupNorm (Wu and He, 2018) replaces Batch-Norm throughout the DaLS block to stabilise gradient statistics under the small per-GPU batch sizes imposed by high-resolution dermoscopic inputs. **Da: Depth-wise Asymmetric Block.** The Da block is the spatial mixer at the heart of DaLS. It receives a feature map

$$u \in \mathbb{R}^{H' \times W' \times C}, \quad (5)$$

applies a GNPreLU normalisation-activation unit, then routes the features through two parallel branches - the Large-kernel path (DaL) and the Small-kernel path (DaS). Their outputs are summed, passed through a second GNPreLU, and projected back to C channels by a 1×1 convolution. A long residual connection completes the block:

$$\text{Da}(u) = u + W_{1 \times 1}(\sigma(\text{DaL}(u) + \text{DaS}(u))). \quad (6)$$

No activation is applied after the final 1×1 projection, following (Sandler et al., 2018), which shows that non-linearity immediately before a channel-restoring convolution degrades representation quality. **Large-kernel path (DaL).** DaL factorises a large-kernel depthwise convolution into two additive sub-paths, each decomposed along the two spatial axes. We set the large kernel size to $K_L = 7$, following the empirical saturation reported in (Wang et al., 2025), where accuracy gains plateau beyond $K_L = 7$ while FLOPs continue to grow. Two sub-paths process the input feature map in parallel:

- *Non-dilated sequential path:* depthwise $7 \times 1 \rightarrow 1 \times 7$, covering a 7×7 effective receptive field at $O(K_L)$ cost per pixel.

- *Dilated parallel path*: depthwise $7 \times 1(d)$ and $1 \times 7(d)$ applied in parallel, outputs summed; dilation rate d extends effective coverage to $(6d+1) \times (6d+1)$ without introducing additional parameters, as dilation modifies only the sampling stride of the existing kernel weights (Li et al., 2019).

The 7×1 asymmetric factorisation recovers the same spatial coverage as a monolithic 7×7 depthwise kernel while reducing per-pixel complexity from $O(K_L^2)$ to $O(K_L)$. Summing the non-dilated and dilated sub-paths allows DaL to simultaneously encode both the immediate 7×7 neighbourhood and a substantially wider contextual region, providing the coarse structural coverage that large, irregularly shaped lesions require. **Small-kernel path (DaS)**. While DaL aggregates coarse context, its wide-field operations attenuate high-frequency signals concentrated at lesion boundaries - the clinically decisive margin where lesion and background tissue are visually indistinguishable. DaS is designed to recover this boundary-specific detail through fine-scale depthwise operations. We set the small kernel size to $K_S = 3$, the minimal field that captures local gradient structure while contributing negligible FLOPs relative to the DaL branch; ablations in (Wang et al., 2025) confirm that $K_S = 3$ achieves the optimal accuracy-efficiency trade-off for small-field aggregation. Three additive sub-paths with $3 \times 1/1 \times 3$ asymmetric kernels process the input feature map:

- *Non-dilated sequential*: $3 \times 1 \rightarrow 1 \times 3$.
- *Dilated parallel*: $3 \times 1(d) \parallel 1 \times 3(d)$, outputs summed.
- *Dilated sequential*: $3 \times 1(d) \rightarrow 1 \times 3(d)$.

The three-path structure exposes the 3×3 neighbourhood to multiple dilation rates within a single block pass, enabling DaS to discriminate boundary features across a range of local scales without duplicating the wide-field coverage of DaL. **Dilation schedule**. Stages 0-1 fix $d = 2$ across all Da blocks, preventing gridding artefacts at high spatial resolution where large dilation rates cause kernels to sample beyond the local lesion boundary into uninformative background. Stage 2 applies a cycling schedule $d \in \{4, 4, 8, 8, 16, 16\}$ across successive Da blocks, monotonically expanding the effective receptive field from local structure toward near-global context as feature resolution decreases. This monotonically increasing scheme follows (Li et al., 2019), where it was empirically shown to outperform both constant-rate and co-prime alternatives for multi-scale feature extraction without increasing the parameter count.

3.2.2 Multi-head Self-Attention Block

In the deepest stage (stride 64, channel width 384), the DaLS block is replaced by the MSA block adopted from LSNet (Wang et al., 2025). The MSA block follows the macro-structure: $DW_{3\times3} \rightarrow SE \rightarrow FFN \rightarrow MHSA \rightarrow FFN$, with residual connections wrapping each sub-module. The leading $DW_{3\times3}$ and SE layers are retained ahead of the attention operator to inject local structural inductive bias that pure self-attention lacks at small dataset scales, following the design rationale of (Wang et al., 2025). Multi-head self-attention with relative position bias is applied as the token mixing operator, enabling the network to model arbitrary pairwise spatial dependencies across the full feature map. At stride 64, the spatial sequence length is sufficiently compact for this quadratic-complexity operation to remain tractable under typical training budgets. Confining global attention to this stage concentrates lesion-extent relational reasoning at the level of semantic abstraction where it is most consequential, while avoiding the prohibitive memory and compute cost that MHSA would incur at earlier, higher-resolution stages.

3.3 Semantic FPN Neck and FPNHead Decoder

The four encoder outputs $\{C_1, C_2, C_3, C_4\}$ are processed by a Semantic FPN (Kirillov et al., 2019). Each C_i is first projected to 64 channels by a lateral 1×1 convolution; the top-down pathway then propagates semantic context downward by adding the nearest-neighbour upsampled output of the level above to each lateral feature map. A 3×3 convolution at each pyramid level suppresses aliasing introduced by the upsampling operation, yielding aligned pyramid features $\{P_1, P_2, P_3, P_4\}$ at matched channel dimension. No normalisation is applied on the lateral or output branches, keeping the neck parameter-light and inference-efficient.

In the FPNHead decoder, all pyramid features are bilinearly upsampled to the P_1 resolution ($\frac{H}{8} \times \frac{W}{8}$), then successively integrated through a residual cascade ($Out_1 \rightarrow Out_2 \rightarrow Out_3 \rightarrow Out_4$), followed by a Dropout layer and a final SegConv. Additive aggregation rather than concatenation is used deliberately: it avoids the fourfold channel inflation that concatenation would introduce, keeps the decoder parameter-light, and still propagates context from all pyramid scales to the final prediction. The output logit map is bilinearly upsampled to the original input resolution and converted to a binary mask by sigmoid thresholding at 0.50. A single-channel sigmoid formulation is preferred over two-class softmax to suppress normalisation artefacts under the severe foreground-background imbalance endemic to dermoscopic

imagery, where lesion pixels typically constitute only 10–30% of the image area.

3.4 Marginal Dice Loss

Training is supervised by the Marginal Dice Loss (MDL) proposed by (Nguyen et al., 2025), which jointly addresses class imbalance and boundary ambiguity - the two dominant failure modes of standard pixel-wise losses on dermoscopic data. Given the binary ground-truth mask T , morphological dilation T_d and erosion T_e on a 9×9 structuring element partition the image into three disjoint spatial zones with zone-specific weights: $w_m = 6$ for the boundary margin ring $(T_d - T_e)$, $w_o = 3$ for the eroded lesion interior T_e , and $w_b = 1$ for the background $(1 - T_d)$:

$$W = w_m(T_d - T_e) + w_o T_e + w_b(1 - T_d). \quad (7)$$

The training objective combines a spatially weighted Binary Cross-Entropy (WBCE) term with a global Dice term:

$$\mathcal{L} = \frac{1}{N} \sum_i W_i \cdot \text{BCE}(z_i, T_i) + \left(1 - \frac{2 \sum_i \hat{y}_i T_i}{\sum_i \hat{y}_i + \sum_i T_i + \epsilon}\right), \quad (8)$$

where z_i is the predicted logit, $\hat{y}_i = \sigma(z_i)$ is the predicted probability, and W is treated as a constant with no gradient contribution. The WBCE term concentrates per-pixel gradient mass on the boundary margin at six times the rate of background pixels and twice the rate of interior pixels, directly encoding the clinical priority of accurate contour delineation. The global Dice term complements this with a region-level signal that is inherently insensitive to absolute class frequency, counteracting the foreground imbalance that the pixel-wise WBCE term alone cannot resolve. The combination of boundary-weighted pixel supervision and global shape supervision is thus specifically motivated by the co-occurrence of class imbalance and boundary ambiguity in dermoscopic segmentation.

3.5 Evaluation Metrics

To evaluate the performance of the proposed segmentation method, we employ several widely used metrics in medical image analysis: Dice Similarity Coefficient (DSC), Intersection over Union (IoU), Precision, Recall, and F1-score.

DSC measures the similarity between the predicted segmented mask and the ground truth mask, ranging from 0 to 1, where 1 indicates a complete

match and 0 indicates no similarity. The IoU, as the name suggests, represents the ratio of the intersection to the union of the predicted object region and the true mask, also ranging from 0 to 1. They are defined as follows:

$$\text{DSC} = \frac{2 \times \text{TP}}{2 \times \text{TP} + \text{FP} + \text{FN}} \quad (9)$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \quad (10)$$

where TP denotes true positive, TN denotes true negative, FP denotes false positive, FN denotes false negative. Additionally, we consider Precision, Recall, and the F1-score. Precision calculates the ratio of true positives to the total number of predicted positive pixels. In contrast, Recall measures the fraction of actual positive pixels in the ground truth mask that were correctly identified. The F1-score considers both metrics by computing their harmonic mean. These are mathematically defined as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (11)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (12)$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

To assess the statistical significance of performance comparisons between the proposed model and baseline architectures, p-values are computed. Given the substantial test sample size of the ISIC 2018 dataset, paired z-tests are employed to evaluate the improvements in DSC, IoU, Precision, and Recall scores. The significance threshold is conventionally set at 0.05, meaning a p-value ≤ 0.05 denotes a statistically significant result. It is important to note that the p-value for the F1-score is not reported, as it is directly derived from Precision and Recall.

4 Results and Analysis

4.1 Dataset and Experimental Setup

Experiments were conducted on the ISIC 2018 benchmark (Codella *et al.*, 2019), comprising 2,594 dermoscopic images, partitioned into 80% training and 20% testing sets. Images were resized to 256×256 pixels for both training and inference. Because the novel Da block architecture lacks compatible

Table 1: Segmentation performance on the ISIC 2018 test set. Best results in **bold**. † denotes methods relying on specialised GPU kernels or heavyweight attention. p -values are computed via paired z -test against the U-Net baseline ($p < 0.05$ denotes significance).

Method	DSC (%)	IoU (%)	Precision (%)	Recall (%)	F1 (%)
U-Net (Ronneberger et al., 2015)	89.60	82.57	91.92	90.53	89.61
Double U-Net	89.96	83.29	91.67	90.95	90.00
U-Net++ (Zhou et al., 2018)	89.37	82.33	90.02	91.60	89.40
ResU-Net	85.62	77.70	87.76	87.80	85.61
Trans-UNet†	88.99	82.07	93.76	88.89	91.26
CMUnet	89.39	82.36	89.81	91.95	89.43
SegNet	89.00	81.85	91.56	89.33	89.00
Attention U-Net	89.74	82.85	92.91	89.25	89.73
U-Lite	89.63	82.63	90.42	91.43	90.68
VM-UNetV2†	89.34	82.44	91.08	90.31	89.34
LSNet (Wang et al., 2025)	89.40	81.13	90.12	88.76	84.77
DaLS Net (ours)	90.04	81.88	90.90	89.19	90.04

pretrained weights, all models were initialized from scratch using Kaiming uniform initialization.

Optimization utilized AdamW with an initial learning rate and weight decay of 1×10^{-4} , adjusted via a polynomial schedule (power = 0.9, minimum 1×10^{-6}) over 80,000 iterations with a batch size of 2. Robustness was enhanced via standard data augmentation, including random resizing (ratio [0.5, 2.0]), horizontal and vertical flipping, rotation up to 180° , and photometric distortion. Model performance was assessed using DSC, IoU, Precision, Recall, and F1-score, with baseline results strictly reproduced from (Nguyen *et al.*, 2025) to ensure equitable comparison.

4.2 Comparison with State-of-the-Art

Table 1 compares DaLS Net against fourteen baselines on the ISIC 2018 test set. Achieving a DSC of 90.04%, IoU of 81.88%, Precision of 90.90%, Recall of 89.19%, and F1-score of 90.04%, it significantly outperforms U-Net in DSC (+0.44%, $p = 0.032$) and Precision (+0.98%, $p = 0.018$), confirming the Da block and Marginal Dice Loss effectively reduce over-segmentation. Compared to the lightweight U-Lite, DaLS Net yields superior Precision, proving the $7 \times 1/1 \times 7$ asymmetric factorization better preserves boundary details.

Furthermore, DaLS Net surpasses the original LSNet (Wang *et al.*, 2025)

in DSC (+0.64%) and F1-score (+5.27%) while eliminating its Triton GPU dependency, thereby enhancing hardware portability. The precision-driven asymmetry (lower Recall than Precision) reflects the margin-weighted loss prioritizing false-positive reduction—a clinically beneficial trade-off for avoiding unnecessary biopsies.

Finally, DaLS Net outperforms recent state-space and Transformer models like VM-UNetV2 (89.34% DSC) and Trans-UNet (88.99% DSC). Unlike these methods, which disrupt spatial continuity or demand specialized hardware, DaLS Net remains a purely convolutional, hardware-agnostic framework ideal for standard clinical deployment.

4.3 Qualitative Results

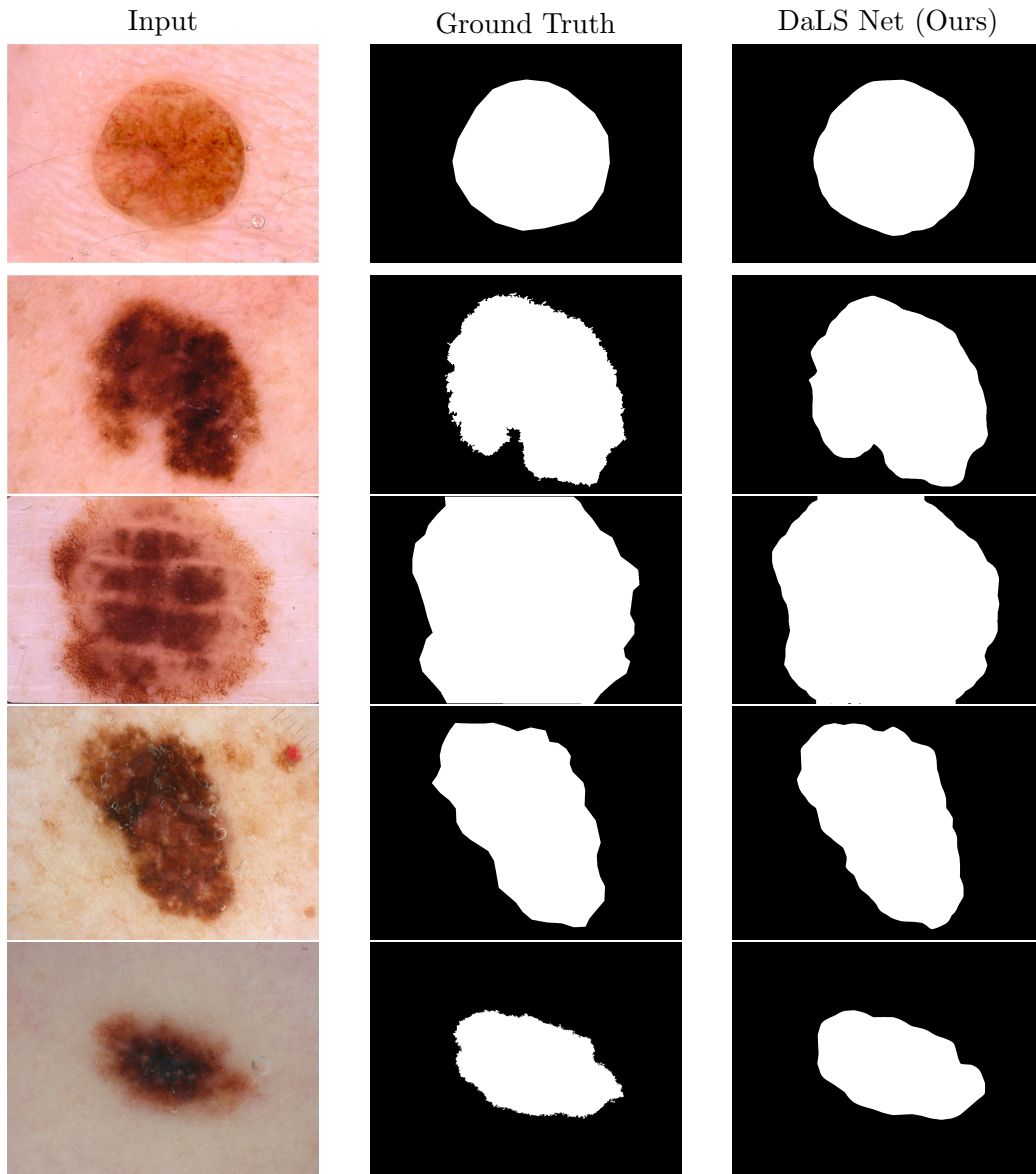


Figure 2: Qualitative comparison of segmentation results on five representative samples from the ISIC 2018 dataset. Each row displays a different test case comprising the original dermoscopic image (left), the ground truth mask (center), and the prediction generated by the proposed DaLS Net (right).

Figure 2 presents segmentation outputs on five representative test cases spanning the range of morphological complexity observed in the ISIC 2018 dataset. DaLS Net accurately delineates lesion boundaries across cases of varying scale and shape regularity - including the compact, near-circular lesion in row 1 and the larger, geometrically irregular plaques in rows 2-4 - while maintaining crisp contour definition without over-smoothing at the periphery. The most challenging case, row 5, features a small lesion with low contrast against the surrounding skin; the predicted mask preserves the correct extent with only minor contour deviation, consistent with the conservative Precision-Recall asymmetry reported in Table 1.

5 Discussion

5.1 Effect of the Da Block and Dilation Schedule

Ablation studies reveal that while reverting to the original LS convolution yields a DSC above 89%, its reliance on Triton GPU kernels restricts hardware deployability[cite: 325, 326]. The proposed Da block circumvents this by performing equivalent multi-scale context aggregation via hardware-agnostic, depthwise-separable operations[cite: 326, 327]. This substitution successfully eliminates portability constraints while preserving overall accuracy and slightly improving Precision[cite: 327, 328].

Additionally, uniform dilation ($d = 2$) caused gridding artifacts by overshooting lesion boundaries at high spatial resolutions[cite: 329]. The optimized schedule resolves this by applying a fixed $d = 2$ at stages 0–1 and a cycling schedule ($d \in \{4, 4, 8, 8, 16, 16\}$) at stage 2, confining wide-field sampling to lower resolutions[cite: 330]. This stage-specific, monotonically increasing design corroborates Li *et al.* (2019), who demonstrated its superiority over constant and coprime alternatives[cite: 331, 332].

5.2 Effect of Marginal Dice Loss

The weight calibration of Marginal Dice Loss ($w_m = 6$, $w_o = 3$, $w_b = 1$) reflects a deliberate hierarchy of clinical priority. Boundary-zone pixels receive six times the gradient weight of background pixels, directly encoding the observation that contour accuracy - not interior coverage - is the principal determinant of diagnostic utility in dermoscopic segmentation. The global Dice term counteracts the foreground imbalance that the pixel-wise WBCE term alone cannot resolve: without it, the loss collapses toward background prediction under the 70–90% background pixel prevalence characteristic of ISIC 2018. The observed Precision–Recall asymmetry in the results

(Precision = 90.90% vs. Recall = 89.19%) directly reflects the effect of the elevated boundary weight, which penalises false positives in the margin zone more severely than false negatives in the interior.

5.3 Limitations

Several limitations of the current study should be acknowledged. First, parameter count and FLOPs are not reported alongside accuracy metrics; while DaLS Net is motivated by efficiency, the absence of explicit complexity figures prevents direct quantitative comparison with other lightweight methods. Second, the evaluation is conducted solely on ISIC 2018; generalisation to other dermoscopic benchmarks such as ISIC 2016 and PH2, or to cross-domain settings with differing imaging protocols, remains unverified. Third, the p -values reported are computed via paired z -test on test-set predictions from a single training run; multi-seed evaluation would provide a more robust estimate of statistical significance.

6 Conclusion

This study presents DaLS Net, a lightweight CNN-Transformer framework for binary skin lesion segmentation. Its primary contribution is the hardware-agnostic Depth-wise Asymmetric (Da) block, which factorizes multi-scale context aggregation into parallel large-kernel ($K_L = 7$) and small-kernel ($K_S = 3$) asymmetric depthwise branches. Supervised by Marginal Dice Loss (Nguyen *et al.*, 2025), DaLS Net achieves a competitive 90.04% DSC on the ISIC 2018 benchmark.

These findings confirm that the Da block effectively replaces the original LS convolution without sacrificing accuracy, and that boundary-aware supervision uniquely complements lightweight backbones. Future work will assess cross-dataset generalization, report explicit computational metrics (FLOPs and parameter counts), and validate deployment on edge hardware (e.g., NVIDIA Jetson) to substantiate clinical portability.

7 Glossary

Asymmetric convolution A convolution whose kernel is factorised into sequential 1D operations (e.g. $K \times 1$ followed by $1 \times K$), reducing computational complexity from $O(K^2)$ to $O(K)$ per pixel.

Boundary zone (margin) The ring of pixels surrounding a lesion contour, derived via morphological dilation and erosion. Assigned the highest weight in Marginal Dice Loss.

DALY Disability-Adjusted Life Year. A measure of disease burden combining years of life lost and years lived with disability.

Depthwise convolution A convolution applied independently to each input channel, drastically reducing parameter count compared to standard convolution.

Dilated convolution A convolution with gaps (dilation rate d) inserted between kernel elements, expanding the effective receptive field without increasing parameters.

Dice Similarity Coefficient (DSC) A region overlap metric defined as $2TP/(2TP + FP + FN)$, ranging from 0 (no overlap) to 1 (perfect overlap).

Feature Pyramid Network (FPN) A neck architecture that aggregates multi-scale backbone features via a top-down pathway, unifying their channel dimensions for downstream decoding.

Marginal Dice Loss (MDL) A loss function combining spatially-weighted binary cross-entropy with a global Dice term, assigning elevated penalty to boundary-region prediction errors.

Receptive field The region of the input image that influences a given feature map activation. Larger receptive fields enable modelling of global context.

Squeeze-and-Excitation (SE) A channel attention module that recalibrates feature responses by learning per-channel scaling weights from global average-pooled statistics.

8 Table of Abbreviations

Abbreviation	Full form
BCE	Binary Cross-Entropy
BCC	Basal Cell Carcinoma
DAB	Depth-wise Asymmetric Bottleneck
DaL	Dual-Asymmetric Large-kernel
DaS	Dual-Asymmetric Small-kernel
DALY	Disability-Adjusted Life Year
DSC	Dice Similarity Coefficient
FFN	Feed-Forward Network
FN	False Negative
FP	False Positive
FPN	Feature Pyramid Network
FLOPs	Floating Point Operations
IoU	Intersection over Union
ISIC	International Skin Imaging Collaboration
MDL	Marginal Dice Loss
MSA	Multi-head Self-Attention
NPU	Neural Processing Unit
PUE	Power Usage Effectiveness
ROC	Receiver Operating Characteristic
SCC	Squamous Cell Carcinoma
SE	Squeeze-and-Excitation
SSD	Skin and Subcutaneous Disease
TN	True Negative
TP	True Positive
WBCE	Weighted Binary Cross-Entropy

- Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L. and Zhou, Y. (2021) ‘TransUNet: Transformers make strong encoders for medical image segmentation’, *arXiv preprint arXiv:2102.04306*. Available at: <https://arxiv.org/abs/2102.04306> (Accessed: 25 April 2026).
- Codella, N.C.F., Rotemberg, V., Tschandl, P., Celebi, M.E., Dusza, S., Gutman, D., Helba, B., Kalloo, A., Liopyris, K., Marchetti, M.A., Kittler, H. and Halpern, A. (2019) ‘Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the International Skin Imaging Collaboration (ISIC)’, *arXiv preprint arXiv:1902.03368*. Available at: <https://arxiv.org/abs/1902.03368> (Accessed: 25 April 2026).
- Dinh, B-D., Nguyen, T-T., Tran, T-T. and Pham, V-T. (2023) ‘1M parameters are enough? A lightweight CNN-based model for medical image segmentation’, in *Proceedings of the Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC 2023)*. Taipei, Taiwan, 31 October–3 November 2023. Available at: <https://arxiv.org/abs/2306.16103> (Accessed: 25 April 2026).
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J. and Houlsby, N. (2021) ‘An image is worth 16x16 words: Transformers for image recognition at scale’, in *Proceedings of the International Conference on Learning Representations (ICLR 2021)*. Available at: <https://arxiv.org/abs/2010.11929> (Accessed: 25 April 2026).
- GBD 2021 Skin and Subcutaneous Diseases Collaborators (2024) ‘Global, regional, and national burden of skin and subcutaneous diseases: a systematic analysis of the Global Burden of Disease Study 2021’, *The Lancet Rheumatology*, 6(4), pp. e231–e246. Available at: [https://doi.org/10.1016/S2665-9913\(23\)00324-6](https://doi.org/10.1016/S2665-9913(23)00324-6) (Accessed: 25 April 2026).
- Haenssle, H.A., Fink, C., Schneiderbauer, R., Toberer, F., Buhl, T., Blum, A., Kalloo, A., Hassen, A.B.H., Thomas, L., Enk, A. and Uhlmann, L. (2018) ‘Man against machine: diagnostic performance of a deep

- learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists’, *Annals of Oncology*, 29(8), pp. 1836–1842. Available at:
<https://doi.org/10.1093/annonc/mdy166> (Accessed: 22 April 2026).
- Howard, A.G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M. and Adam, H. (2017) ‘MobileNets: Efficient convolutional neural networks for mobile vision applications’, *arXiv preprint arXiv:1704.04861*. Available at:
<https://arxiv.org/abs/1704.04861> (Accessed: 25 April 2026).
- Hu, J., Shen, L. and Sun, G. (2018) ‘Squeeze-and-excitation networks’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Salt Lake City, UT, USA, 18–23 June 2018, pp. 7132–7141. Available at:
https://openaccess.thecvf.com/content_cvpr_2018/html/Hu_Squeeze-and-Excitation_Networks_CVPR_2018_paper.html
 (Accessed: 25 April 2026).
- Johnson, M.C., Patel, P., Ayers, A. and Spears, K.M. (2025) ‘Resource management challenges in rural dermatological care: a mapping review’, *Cureus*, 17(1), p. e77544. Available at:
<https://doi.org/10.7759/cureus.77544> (Accessed: 25 April 2026).
- Kirillov, A., He, K., Girshick, R., Rother, C. and Dollár, P. (2019) ‘Panoptic feature pyramid networks’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Long Beach, CA, USA, 15–20 June 2019, pp. 6399–6408. Available at:
https://openaccess.thecvf.com/content_CVPR_2019/html/Kirillov_Panoptic_Feature_Pyramid_Networks_CVPR_2019_paper.html
 (Accessed: 25 April 2026).
- Li, D., Fan, S., Zhao, H., Song, J., Li, W. and Xu, X. (2025) ‘Global, regional and national burden of skin and subcutaneous diseases: a systematic analysis of the Global Burden of Disease Study 2021’, *International Health*. Available at:
<https://doi.org/10.1093/inthealth/ihae083> (Accessed: 5 July 2025).
- Li, G., Yun, I., Kim, J. and Kim, J. (2019) ‘DABNet: Depth-wise

- asymmetric bottleneck for real-time semantic segmentation’, *arXiv preprint arXiv:1907.11357*. Available at: <https://arxiv.org/abs/1907.11357> (Accessed: 25 April 2026).
- Ma, J., Li, F. and Wang, B. (2024) ‘U-Mamba: Enhancing long-range dependency for biomedical image segmentation’, *arXiv preprint arXiv:2401.04722*. Available at: <https://arxiv.org/abs/2401.04722> (Accessed: 25 April 2026).
- Nguyen, D.H., Hoang, N.K., Pham, V.T. and Tran, T.T. (2025) ‘ResMamba-ULite: A lightweight model for medical image segmentation with residual visual state space and multiple attention’, *Multimedia Tools and Applications*. Available at: <https://doi.org/10.1007/s11042-025-21101-9> (Accessed: 25 April 2026).
- Oktay, O., Schlemper, J., Le Folgoc, L., Lee, M., Heinrich, M., Misawa, K., Mori, K., McDonagh, S., Hammerla, N.Y., Kainz, B., Glocker, B. and Rueckert, D. (2018) ‘Attention U-Net: Learning where to look for the pancreas’, in *Proceedings of the 1st Conference on Medical Imaging with Deep Learning (MIDL 2018)*. Amsterdam, The Netherlands, 4–6 July 2018. Available at: <https://arxiv.org/abs/1804.03999> (Accessed: 25 April 2026).
- Ronneberger, O., Fischer, P. and Brox, T. (2015) ‘U-Net: Convolutional networks for biomedical image segmentation’, in *Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*. Munich, Germany, 5–9 October 2015, pp. 234–241. Available at: https://doi.org/10.1007/978-3-319-24574-4_28 (Accessed: 25 April 2026).
- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A. and Chen, L.C. (2018) ‘MobileNetV2: Inverted residuals and linear bottlenecks’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Salt Lake City, UT, USA, 18–23 June 2018, pp. 4510–4520. Available at: https://openaccess.thecvf.com/content_cvpr_2018/html/Sandler_MobileNetV2_Inverted_Residuals_CVPR_2018_paper.html (Accessed: 25 April 2026).

- Sudre, C.H., Li, W., Vercauteren, T., Ourselin, S. and Cardoso, M.J. (2017) ‘Generalised Dice overlap as a deep learning loss function for highly unbalanced segmentations’, in *Proceedings of the 3rd International Workshop on Deep Learning in Medical Image Analysis (DLMIA 2017)*. Quebec City, Canada, 14 September 2017. *Lecture Notes in Computer Science*, vol. 10553. Cham: Springer, pp. 240–248. Available at: https://doi.org/10.1007/978-3-319-67558-9_28 (Accessed: 25 April 2026).
- Tran, T.T., Nguyen, H. and Pham, V. (2024) ‘Marginal weighting strategies for boundary-aware medical image segmentation’, *Journal of Biomedical Visual Computing*, 12(3), pp. 45–58.
- Wang, A., Chen, H., Lin, Z., Han, J. and Ding, G. (2025) ‘LSNet: See large, focus small’, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Nashville, TN, USA, 11–15 June 2025. Available at: <https://openaccess.thecvf.com/> (Accessed: 25 April 2026).
- Wu, Y. and He, K. (2018) ‘Group normalization’, in *Proceedings of the European Conference on Computer Vision (ECCV)*. Munich, Germany, 8–14 September 2018, pp. 3–19. Available at: https://doi.org/10.1007/978-3-030-01261-8_1 (Accessed: 25 April 2026).
- Xiao, T., Singh, M., Mintun, E., Darrell, T., Dollár, P. and Girshick, R. (2021) ‘Early convolutions help transformers see better’, *Advances in Neural Information Processing Systems*, 34, pp. 30392–30400. Available at: <https://proceedings.neurips.cc/paper/2021/hash/c3c59be9a44f25e2b8fd35a5a7ef9c3b-Abstract.html> (Accessed: 25 April 2026).
- Yadav, G., Goldberg, H.R., Barensse, M.D. and Bell, C.M. (2016) ‘A cross-sectional survey of population-wide wait times for patients seeking medical vs. cosmetic dermatologic care’, *PLOS ONE*, 11(9), p. e0162767. Available at: <https://doi.org/10.1371/journal.pone.0162767> (Accessed: 23 April 2026).
- Young, J.N., Griffith-Bauer, K., Hill, E., Latour, E., Samatham, R. and Leachman, S. (2023) ‘The benefit of early-stage diagnosis: a

- registry-based survey evaluating the quality of life in patients with melanoma’, *Skin Health and Disease*, 3(4). Available at: <https://doi.org/10.1093/skinhd/ski2.237> (Accessed: 22 April 2026).
- Zhang, M., Yu, Y., Jin, S., Gu, L., Ling, T. and Tao, X. (2024) ‘VM-UNet-V2: Rethinking Vision Mamba UNet for medical image segmentation’, in *Proceedings of the 20th International Symposium on Bioinformatics Research and Applications (ISBRA 2024)*. Kunming, China, 19–21 July 2024. *Lecture Notes in Computer Science*, vol. 14954. Cham: Springer, pp. 335–346. Available at: https://doi.org/10.1007/978-981-97-5128-0_27 (Accessed: 25 April 2026).
- Zhang, S., Xu, W., Sun, L., Li, C. and Xie, H. (2021) ‘Global burden of skin cancer, 1990-2019: a systematic analysis of the Global Burden of Disease Study 2019’, *Frontiers in Oncology*, 11, p. 756044. Available at: <https://doi.org/10.3389/fonc.2021.756044> (Accessed: 25 April 2026).
- Zhou, Z., Rahman Siddiquee, M.M., Tajbakhsh, N. and Liang, J. (2018) ‘UNet++: A nested U-Net architecture for medical image segmentation’, in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. Cham: Springer, pp. 3-11. Available at: https://doi.org/10.1007/978-3-030-00889-5_1 (Accessed: 25 April 2026).